TalentSyncAI

TalentSyncAI is an AI-powered platform with two independent experiences, determined by how each user registers: as a candidate or as a recruiter. Candidates work on their own career goals, with CV-to-job matching, skill-gap analysis, company and role research, tailored resumes, and personalized interview preparation. Recruiters work inside their company's private workspace, where they upload and manage a CV pool, search it in natural language, and receive ranked candidates with match scores and clear explanations of strengths and gaps.

Team Members

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Proposal

Features divided across Frontend, Backend (.NET), and AI teams

Frontend Team	Backend Team (.NET)	AI Team
Candidate and recruiter interfaces, dashboards, search/chat, results, workspace and file previews.	APIs, authentication, multi-tenant company data, CV storage, runs, search orchestration, security and infrastructure.	CV understanding, requirement extraction, semantic retrieval, matching, ranking, explanations and career agents.

1. Frontend Team

Candidate Side

- Registration/login and protected user experience.
- Create and manage target job applications.
- CV upload and Job Description input by text, file, or URL.
- AI chat with streaming progress and multi-turn editing.
- Candidate-job match and skill-gap presentation.
- Company/role research presentation.
- Tailored resume viewer and PDF download.
- Interview preparation, interview rounds, questions, self-introduction and STAR stories.
- Personalized study/preparation plan and interview battlecard.
- Workspace and preview for uploaded/generated files.

Company / Recruiter

- Company/recruiter login and dedicated recruitment dashboard.
- Upload, browse and manage the company's CV/candidate pool.
- Natural-language candidate search box, e.g. 'Find .NET developers with 3+ years, PostgreSQL and Docker'.
- Optional structured filters such as skills, years of experience, role and other approved criteria.
- Ranked candidate results with match score.
- Show why each candidate matched: strengths, matching requirements and gaps.
- Candidate profile/details page.
- Shortlist, reject/save candidate actions and saved searches.
- Candidate comparison view for selected CVs.

Settings

- Model/provider selection UI when enabled.
- User/company workspace preferences.

2. Backend Team (.NET)

Core API & Security

- ASP.NET Core REST API consumed by the frontend.
- OpenAPI/Swagger contracts, validation and consistent errors.
- Authentication, authorization and session/token handling.
- Candidate and recruiter/company roles and permissions.
- Strict multi-tenant isolation: one company cannot access another company's CVs or searches.

Candidate Application Backend

- Applications, CV/JD uploads, processing runs and results.
- PostgreSQL persistence for users, applications, runs and metadata.
- S3-compatible storage for uploaded documents and generated PDFs.
- Background processing, retries, cancellation and run-stage tracking.
- SignalR/SSE-style live progress to the frontend.

Company / Recruitment Backend

- Company, recruiter, candidate and CV entities/endpoints.
- Bulk/single CV upload and file metadata management.
- Persist normalized candidate profiles extracted from CVs.
- Search endpoint accepting natural-language requirements and optional filters.
- Orchestrate retrieval and AI ranking while keeping the backend as source of truth.
- Persist search history, ranking results, shortlists and recruiter actions.
- Pagination/filtering for large candidate pools.
- Use PostgreSQL + pgvector/vector store where selected for semantic candidate retrieval.
- Do not send the entire CV database to the AI for every query; retrieve a relevant candidate subset first.

AI Integration

- Internal HTTP/JSON contract with the AI service.
- Send run/application/search identifiers plus only required data.
- Receive and validate structured AI results against DTO/JSON schemas.
- Support asynchronous run IDs, polling/callbacks, or queues for long operations.

Operations

- Logging, observability and health checks.
- Secrets/configuration management.
- Dockerized service networking and deployment.

3. AI Team

Candidate Career Agents

- Parse and understand CV and Job Description content.
- Company and role research.
- Candidate-job matching and gap analysis.
- Resume tailoring without inventing candidate facts.
- Interview preparation, questions, self-introduction and STAR stories.
- Study plan and battlecard generation.
- Multi-agent orchestration, tool calling, structured outputs and memory where in scope.

Company Candidate Intelligence

- Parse every uploaded company CV into a normalized candidate profile.
- Extract searchable information such as skills, roles, experience and education from CV content.
- Generate embeddings/indexable representations for semantic retrieval where selected.
- Understand recruiter questions written in natural language.
- Convert recruiter questions into structured search requirements.
- Retrieve/match candidates against the requested requirements.
- Re-rank the most relevant candidates.
- Produce a match score/ranking and an explanation grounded in each CV.

- Identify matching strengths and missing requirements for each result.
- Support follow-up recruiter queries/refinements without requiring the entire candidate pool to be reprocessed.

AI Safety / Quality

- Ground recommendations in CV content and avoid fabricated experience.
- Return structured machine-readable outputs to the backend.
- Keep scoring criteria explainable and consistent enough for recruiter review.
- AI ranking should support human decision-making rather than automatically making final hiring decisions.

Company-Side Feature Flow

1. Recruiter uploads CVs -> .NET Backend -> Object Storage.
2. Backend requests AI parsing -> structured candidate profile + optional embedding/index data.
3. Backend persists candidate profile and search index/vector reference under the correct company tenant.
4. Recruiter asks a natural-language question describing required skills/experience.
5. Backend + search layer retrieves the most relevant candidate subset.
6. AI re-ranks candidates and returns match scores, strengths, gaps and explanations.
7. Backend validates/saves results -> Frontend displays ranked candidates for recruiter review.

Overall Product Structure

Candidate Side	**Company / Recruiter Side**
CV + Job Description analysis	Company CV repository
Company and role research	Natural-language candidate search
Job-fit / skill-gap analysis	Semantic candidate retrieval
AI-tailored resume	AI matching and ranking
Interview preparation + STAR stories	Match explanation, strengths and gaps
Study plan + battlecard	Shortlist, saved searches and candidate comparison